

SOFTWARE PROJECT MANAGEMENT

WHAT IS SOFTWARE PROJECT MANAGEMENT?

- ❖ Software project management is an art and discipline of planning and supervising software projects. It is a sub-discipline of software project management in which software projects planned, implemented, monitored and controlled.
- ❖ It is a procedure of managing, allocating and timing resources to develop computer software that fulfills requirements.
- ❖ In software Project Management, the client and the developers need to know the length, period and cost of the project.

WHY IS SOFTWARE PROJECT MANAGEMENT IMPORTANT?

- Strategic Alignment
- Leadership
- Clear Focus & Objectives
- Realistic Project Planning
- Quality Control
- Risk Management
- Managing and Learning from Success and Failure

ROLES AND RESPONSIBILITY

- ❖ **Planning:** This means putting together the blueprint for the entire project from ideation to fruition. It will define the scope, allocate necessary resources, propose the timeline, delineate the plan for execution, lay out a communication strategy, and indicate the steps necessary for testing and maintenance.
- ❖ **Leading:** A software project manager will need to assemble and lead the project team, which likely will consist of developers, analysts, testers, graphic designers, and technical writers. This requires excellent communication, people and leadership skills.

ROLES AND RESPONSIBILITY

- ❖ **Execution:** The project manager will participate in and supervise the successful execution of each stage of the project. This includes monitoring progress, frequent team check-ins and creating status reports.
- ❖ **Time management:** Staying on schedule is crucial to the successful completion of any project, but it's particularly challenging when it comes to managing software projects because changes to the original plan are almost certain to occur as the project evolves. Software project managers must be experts in risk management and contingency planning to ensure forward progress when roadblocks or changes occur.

ROLES AND RESPONSIBILITY

- ❖ **Budget:** Like traditional project managers, software project managers are tasked with creating a budget for a project, and then sticking to it as closely as possible, moderating spend and re-allocating funds when necessary.
- ❖ **Maintenance:** Software project management typically encourages constant product testing in order to discover and fix bugs early, adjust the end product to the customer's needs, and keep the project on target. The software project manager is responsible for ensuring proper and consistent testing, evaluation and fixes are being made.

ASPECTS OF PROJECT MANAGEMENT SOFTWARE

- ❖ Issue tracking,
- ❖ Risk management,
- ❖ Time tracking,
- ❖ Project budgeting and
- ❖ Cost management,
- ❖ Resources planning



NEED OF SOFTWARE PROJECT MANAGEMENT

- ❖ Time
- ❖ Cost
- ❖ Quality



The 5 Phases of Project Management

Project Life Cycle

The 5 Phases of Project Management is also called the **Project Lifecycle**. The 5 Phases of Project Management consist of:

- ☐ Project Initiation Phase
- ☐ Project Planning Phase
- ☐ Execution Phase
- ☐ Monitoring and Control Phase
- ☐ Project Closeout Phase



The 5 Phases of Project Management

Initiation Phase

- ❑ The beginning phase of every project where a project manager is selected.
- ❑ The project manager will meet with the project sponsor and key stakeholders to determine the business or technical objectives of the project.
- ❑ Determine any historical information regarding the project. Begin to identify who is needed on a project team.

Key Outputs of this phase are:

- ❑ Develop Project Charter
- ❑ Develop Preliminary Scope Statement
- ❑ Identification of all stakeholders
- ❑ Divide the project into Phases



The 5 Phases of Project Management

Planning Phase

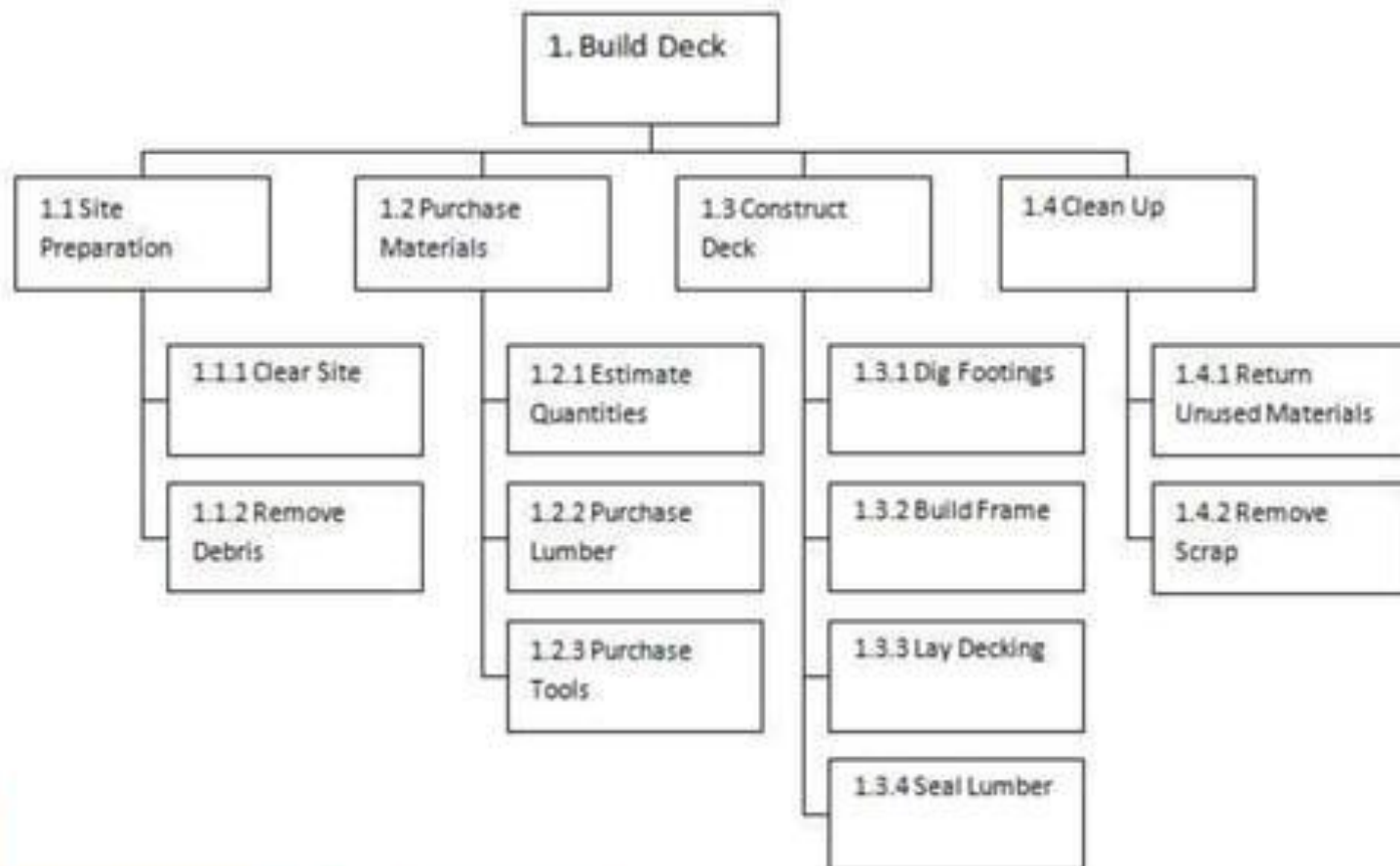
- ❑ One of the largest and most important phases of every project. The project manager will determine the team, identify roles and responsibilities and solidify a scope statement.
- ❑ At this phase the project manager will begin to develop different project plans.

They consist of :

- ❑ Communication Plan (How will you communicate to your team, to your stakeholders?)
- ❑ Risk Plan (How will risks be documented how will they be escalated?)
- ❑ Testing Plan
- ❑ Project Schedule (Timeline)
- ❑ Develop a WBS – Work Break Down Structure A work break down structure is key because it breaks down the work into manageable work activities and/or tasks.

The 5 Phases of Project Management

Work Break Down Structure



The 5 Phases of Project Management

Planning Phase

The number of outputs for the planning phase is large and that is why the planning phase should be given adequate time for planning.

*Please keep in mind that this phase will have is an iterative process. Everything will be revised until all parties agree on the outputs and formal approval is given.



Triple Constraint

Is the methodology that if there is a change in cost, then the change will impact schedule or the scope and vice versa.

The 5 Phases of Project Management

Planning Phase

Key Outputs of this phase are:

- ❑ Create WBS and WBS Dictionary
- ❑ Develop Schedule
- ❑ Develop Budget
- ❑ Determine roles and Responsibilities
- ❑ Develop Communication Plan
- ❑ Develop Risk Plan
- ❑ Create Performance measurement baselines
- ❑ Develop Preliminary Scope Statement
- ❑ Test Plan
- ❑ Determine the Critical Path
- ❑ Prepare procurement documents
- ❑ Create Process Improvement Plan
- ❑ Gain Formal Approval of all plans and key deliverables

* After that you as the project manager are good to Launch Kickoff Meeting!

The 5 Phases of Project Management

Execution Phase

- ❑ This phase is where you do the “work” or execute the plan. This phase typically requires the most amount of time since the majority of the work is done in this timeframe.
- ❑ You will solidify the team; conduct teambuilding exercises.
- ❑ Hold progress and status meetings
- ❑ Follow the processes in your project plans.
- ❑ Recommend changes and corrective actions.
- ❑ Continuous Improvement

Key outputs of this phase are:

- ❑ Signoff that key milestones are complete.
- ❑ Documented performance reports



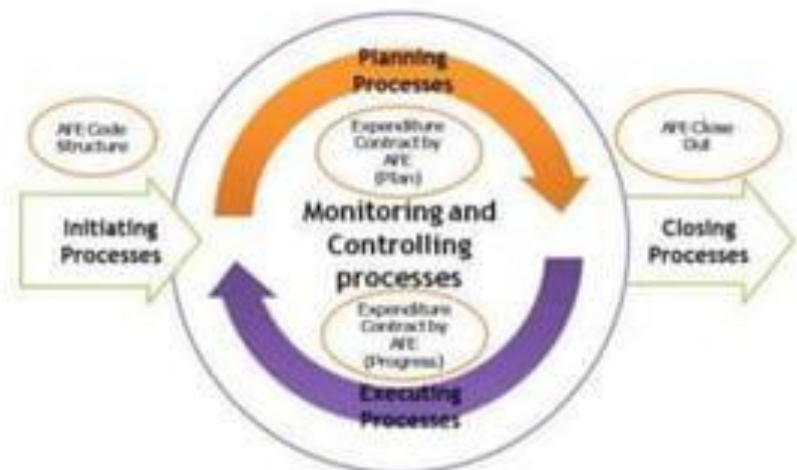
The 5 Phases of Project Management

Monitor and Controlling Phase

- ❑ The phase where results derived from your new process or product are compared to the metrics you outlined in your performance plan.
- ❑ Implement your change control process
- ❑ Determine variances and identify if they warrant corrective action or change
- ❑ Facilitate conflict Resolution
- ❑ Use Issue Logs

Key outputs of this phase are:

- ❑ Receive all signoffs that work is completed as per requirements.
- ❑ Archive all project assets
- ❑ Create and document lessons learned



The 5 Phases of Project Management

Project Closure Phase

This is the final phase of the project life cycle.

- ❑ Develop closure procedures.
- ❑ Document final performance reporting
- ❑ Handoff completed product, process or code.
- ❑ Release resources
- ❑ Complete contract closures.

Key outputs of this phase are:

- ❑ Receive all signoffs that work is completed as per requirements.
- ❑ Index and archive all project assets
- ❑ Create and document lessons learned



The 5 Phases of Project Management

Celebrate project success!

